

# Code Sect 5 - CMAN Ian Wyse

Task 1:

$$PE: \begin{pmatrix} PE(0)(0) = 5 & PE(0)(1) = 6 \\ PE(1)(0) = 7 & PE(1)(1) = 8 \end{pmatrix}$$

Task 2:

$$C0 \quad \begin{matrix} 3 & 1 \\ 4 & 2 \end{matrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

$$C1 \quad \begin{matrix} 3 & 1 \\ 4 & 2 \end{matrix} \begin{pmatrix} 5 & 1 \cdot 5 & 6 \\ 7 & & 8 \end{pmatrix}$$

$$C2 \quad \begin{matrix} 3 & 1 \\ 4 & 2 \end{matrix} \begin{pmatrix} 5 & 3 \cdot 5 & 6 & 1 \cdot 6 \\ 7 & 2 \cdot 7 + 5 & 8 & \end{pmatrix}$$

$$C3 \quad \begin{matrix} 3 & 1 \\ 4 & 2 \end{matrix} \begin{pmatrix} 5 & 3 \cdot 6 \\ 7 & 4 \cdot 7 + 15 & 2 \cdot 8 + 6 \end{pmatrix}$$

$$C4 \quad \begin{matrix} 3 & 1 \\ 4 & 2 \end{matrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 & 4 \cdot 8 + 18 \end{pmatrix}$$

$$C5 \quad \begin{matrix} 3 & 1 \\ 4 & 2 \end{matrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

Final Result:

$$\begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Table:

|    | RO <sub>in</sub> | R1 <sub>in</sub> | P <sub>00</sub> | P <sub>01</sub> | P <sub>10</sub> | P <sub>11</sub> | Outputs                                   |
|----|------------------|------------------|-----------------|-----------------|-----------------|-----------------|-------------------------------------------|
| C0 | -                | -                | -               | -               | -               | -               | -                                         |
| C1 | 1                | -                | 5               | -               | -               | -               | -                                         |
| C2 | 3                | 2                | 15              | 6               | 19              | -               | -                                         |
| C3 | -                | 4                | -               | 18              | 43              | 22              | C <sub>00</sub> = 19                      |
| C4 | -                | -                | -               | -               | -               | 50              | C <sub>01</sub> = 22 C <sub>10</sub> = 43 |
| C5 | -                | -                | -               | -               | -               | -               | C <sub>11</sub> = 50                      |

Task 3:

- A total of 8 MAC operations were performed
- Each input value was reused once. (Was used 2 total times)
- There was one access for each input and output value, for a total of 12 memory accesses.

Task 4:

If this were output stationary, the partial sums would remain stationary and the values from matrix B would flow down through the systolic array